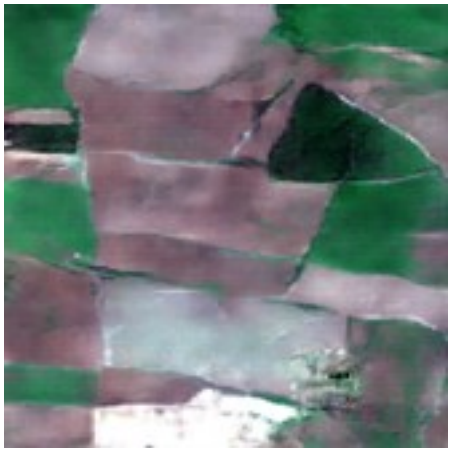
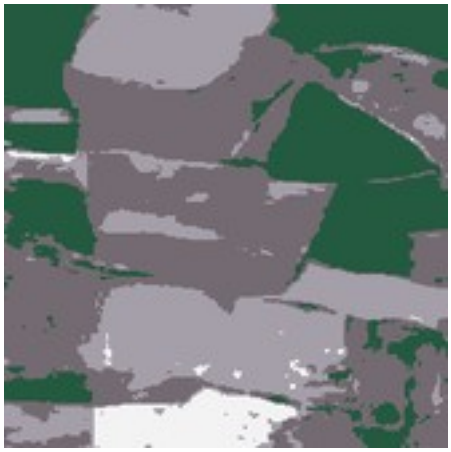
SAR Image Analysis Report

Original SAR



Colorized



K-Means Segmented

Metadata:

- Original File: original_1764518284.png
- Colorized File: colorized_1764518284.png
- Processed On: 2025-11-30 21:28:45 IST
- SAR Classifier Confidence: 0.994
- Colorization Model: Pix2Pix GAN
- KMeans Clusters: 4
- Colorized Resolution: 256 x 256 px

Cluster Analysis:

Cluster 0: Likely Soil / Barren Land (Reddish Tone) | Mean RGB: (116, 106, 113) | Area: 42.61%

Cluster 1: Likely Vegetation (Green Tone) | Mean RGB: (36, 90, 64) | Area: 26.75%

Cluster 2: Likely Water (Blue Tone) | Mean RGB: (164, 160, 170) | Area: 26.02%

Cluster 3: Likely Water (Blue Tone) | Mean RGB: (243, 243, 243) | Area: 4.62%

Land Cover Summary:

- Likely Soil / Barren Land (Reddish Tone) → 42.61%
- Likely Vegetation (Green Tone) → 26.75%
- Likely Water (Blue Tone) → 26.02%
- Likely Water (Blue Tone) → 4.62%

Note: Colorized image is GAN-generated and for visualization only.

Analysis Details & Recommendations

Interpretation Notes:

The area is dominated by likely soil / barren land (reddish tone) (42.61% combined).

Secondary presence: likely water (blue tone) (30.64%).

Secondary presence: likely vegetation (green tone) (26.75%).

K-Means groups pixels by color similarity — labels are heuristic.

Consider adding multispectral or time-series data for improved analysis.

Recommendations:

The analysis shows dominant land cover is soil/barren (42.59%). Large soil/barren area suggests land degradation or exposed surfaces; consider erosion control or land restoration planning. The second-largest class is vegetation (26.67%), so combined management strategies addressing both soil/barren and vegetation are advisable. Recommended actions: collect ground-truth samples where possible, prioritize monitoring based on dominant class, and re-run analysis after incorporating higher-resolution data or additional bands.

Confidence Indicators:

SAR Detection Confidence: 0.994
Dominant Class Percentage: 42.61%
Classification Reliability: High

Limitations:

- KMeans is unsupervised and may not match exact land-cover classes.
- GAN colorization is synthetic and does not represent true spectral values.
- Classification depends on image quality and may contain uncertainty.
- Single-image analysis; use multi-temporal data for trend-based conclusions.

Final Summary:

The analysis indicates predominant vegetation with mixed surfaces.

No major water bodies or dense urban zones detected in this image.

Recommend periodic monitoring and ground-truth sampling to improve confidence.